

Simple Protection for Audio Amplifier Loudspeakers

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Most audio power amplifiers have non-regulated power supply and, at some moments, their output power can be much higher than the specified limits. This can damage the loudspeakers if they are not protected against high input power. Also, during experiments with audio power amplifiers, the output power can be higher than expected.

Consequently, it is desirable to have a protection block between the audio power amplifier and loudspeaker, which can reduce the higher power applied to the loudspeaker in order to protect it from any damage.

Presented here is a simple protection circuit to protect the loudspeaker from excess power output of the audio

power amplifier. The circuit is connected between the power amplifier output and the loudspeaker. In case of overload, the circuit connects additional resistors between the load and the power amplifier, reducing the power to the load.

Circuit and working

The protection circuit for the loudspeaker of an audio power amplifier is shown in Fig. 1. The signal from the audio power amplifier is applied to connector CON1. The loudspeaker is connected at CON3. The signal from CON1 is rectified by bridge rectifier BR1. Filtering capacitor C1 and resistor R1 determine the contact time of the unit. The value of capacitor C1 can be reduced or increased to meet the particular requirements.

The circuit has two identical stages switching on and off protection resistors through electromagnetic relays. The number of stages can be changed in order to get the required effect. The voltage level for activation of each stage can be adjusted using potentiometers POT1 and POT2. If adjustment is not

needed, these presets can be replaced with fixed-value resistors.

The circuit can take power supply from the power amplifier connected to CON1 or from the DC power source applied to connector CON2. In the first case, switch S1 is closed and external power supply should not be applied to connector CON2. In the second case, switch S1 is open and external power supply in the range of 12-16V should be applied to connector CON2. For voltage level adjustment using potentiometers POT1 and POT2, appropriate well-regulated external power supply should be used. Voltage levels can be readjusted according to your requirements.

Transistors T1 and T2 can be any low-power transistors with gain higher than 400; e.g., BC550C and BC108C. LED1 provides light proportional to the input signal. The maximum current for LED1 can be adjusted using an appropriate value of resistor R6.

Each of relays RL1 and RL2 should have at least one normally-closed (N/C) contact, which is used to connect/disconnect protection resistors R4 and R5 in series with loudspeaker LS1.

Working of the circuit is simple. When the input signal is lower than both the thresholds, the N/C relay contacts are closed and loudspeaker LS1 is directly connected to connector CON1. Transistor T1 is set at lower threshold using POT1 and transistor T2 is set at higher threshold using POT2.

Test Points

Test point	Details
TP0	0V, GND
TP1	12V with switch S1 closed and no audio applied
TP2	Threshold voltage low
TP3	Threshold voltage high

PARTS LIST

Semiconductors:

- T1, T2 - BC547 npn transistor
- LED1 - 5mm LED
- BR1 - 1A bridge rectifier
- D1, D2 - 1N4001 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1 - 12-ohm
- R2, R3 - 4.7-kilo-ohm
- R4, R5 - 8-ohm, 25W
- R6 - 2-kilo-ohm
- POT1, POT2 - 2.7-kilo-ohm potentiometer

Capacitors:

- C1 - 100 μ F, 35V electrolytic

Miscellaneous:

- CON1 - 2-pin connector for audio input
- CON2 - 2-pin connector for 12V battery
- LS1 - 8-ohm, 50W loudspeaker
- S1 - On/off toggle switch
- RL1, RL2 - 12V, 1C/O relays

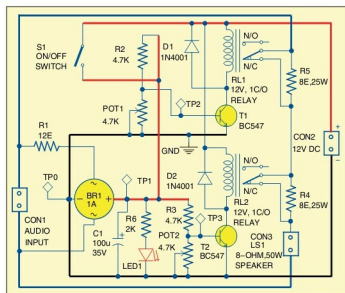


Fig. 1: Simple protection circuit for the loudspeakers of an audio amplifier